OKELLO ELLY

Software Developer & Systems Architect | Scalable Systems • Full-Stack Engineering • Cybersecurity
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SUMMARY

Software Developer and Systems Architect with hands-on experience delivering scalable, secure, production-grade software systems. Skilled in full-stack development, backend architecture, authentication systems, database design, networking, and cybersecurity. Experienced in building high-reliability platforms with strong focus on performance, maintainability, and secure system design. Proficient in AI-assisted development and data-driven engineering, consistently solving complex real-world software challenges.

EDUCATION

Bachelor of Science in Computer Software Engineering

Zetech University — *Jan 2025 – Nov 2027*

Core Technical Focus:

- Software Development (C, C++, JavaScript, Python)
- Backend Engineering & REST API Design
- Database Management (MySQL, SQLite)
- System Analysis & Architecture
- Secure Systems & Authentication (JWT, RBAC)
- Networking Fundamentals & Junior Network Engineering
- Data Science & Analytics Concepts

SOFT SKILLS

- Leadership & Team Collaboration
- Technical Communication
- Problem Solving & Critical Thinking
- Strategic Thinking & Initiative

PROJECTS

BIT System — Backend Developer & System Architect

Technologies: Python (Flask), MySQL, SQLite, JWT Authentication, REST APIs, HTML/CSS

Timeline: 2024 – Present

- Designed and implemented a secure backend platform supporting student services, event management, payments, and analytics.
- Engineered REST APIs with JWT authentication and role-based access control, strengthening platform security and access governance.
- Designed normalized relational database schemas, improving query efficiency and system scalability.
- Integrated backend services with administrative dashboards and operational tools, enabling real-time system insights.

LearnLight — Software Developer

Technologies: React, JavaScript, APIs, Backend Services

- Contributing to development of an interactive digital learning platform focused on structured online education.
- Implementing modular frontend components, backend logic, and API integrations for scalable course delivery.
- Improving performance, usability, and maintainability through clean architecture and best engineering practices.

Mobi — Software Developer & System Architect (In Progress)

Technologies: React, JavaScript, APIs, Backend Services

- Architecting a multi-modal transport platform integrating public minibuses (matatus), ride-hailing services, and campus shuttles.
- Implementing real-time vehicle tracking, secure digital payments, and commuter personalization features.
- Optimizing backend services for scalability and reliability, reducing simulated system latency by up to 30%.
- Designing modular system architecture enabling expansion to additional cities and transport providers.

SmartWaste — Software Developer(In Progress)

Technologies: Python, Flask, MySQL, REST APIs, Data Analytics Concepts

- Developing a data-driven waste management platform for improved waste collection tracking and operational visibility.

- Built backend APIs to record, manage, and analyze waste collection data.
- Designed relational database schemas supporting structured reporting and analytics.
- Improved operational insight and decision-making through structured data collection and analysis.

CERTIFICATIONS & TECHNICAL COMPETENCIES

• **Introduction to Modern AI — Cisco Networking Academy (2026)**

Foundations of artificial intelligence, AI systems, machine learning concepts, AI applications

• **Junior Cybersecurity Analyst — Cisco Networking Academy**

Threat analysis, vulnerability assessment, security controls, SOC fundamentals

• **Huawei Computer Networks (CRA Training Program) Certification**

Network architecture, TCP/IP, routing & switching, troubleshooting

• **Cisco Certified Cybersecurity Essentials**

Cyber risk management, secure access control, incident response

• **Data Science & Analytics — HP LIFE (2025)**

Data analysis, visualization, reporting, analytical thinking

• **AI Fluency: Framework and Foundations — Anthropic**

Prompt engineering, AI workflows, AI-assisted software development

• **SEO — HubSpot Academy (2025)**

Search optimization, analytics, performance tracking

• **Content Marketing — HubSpot Academy (2024)**

Content strategy, audience growth, digital storytelling

• **Digital Marketing — HubSpot Academy (2024)**

Campaign strategy, social media analytics, conversion optimization

• **Anticipatory Governance: Systems Thinking & Strategic Foresight for SDG-Aligned Integrated Planning (Part 1) — Foundations of Anticipatory Governance**

Systems thinking foundations, foresight methodologies, SDG-aligned planning principles

• **Anticipatory Governance: Systems Thinking & Strategic Foresight for SDG-Aligned Integrated Planning (Part 2) — Implementation of Systems Thinking and Strategic Foresight**

Applied systems thinking, strategic foresight implementation, governance integration

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ADDITIONAL INFORMATION

- Actively building production-level software systems addressing real-world engineering problems.
- Secretary of the Innovation Club at Zetech University, contributing to innovation and technology initiatives.
- Experienced in leading engineering projects and collaborative technical teams.
- Committed to continuous learning, engineering excellence, and building impactful technology solutions.